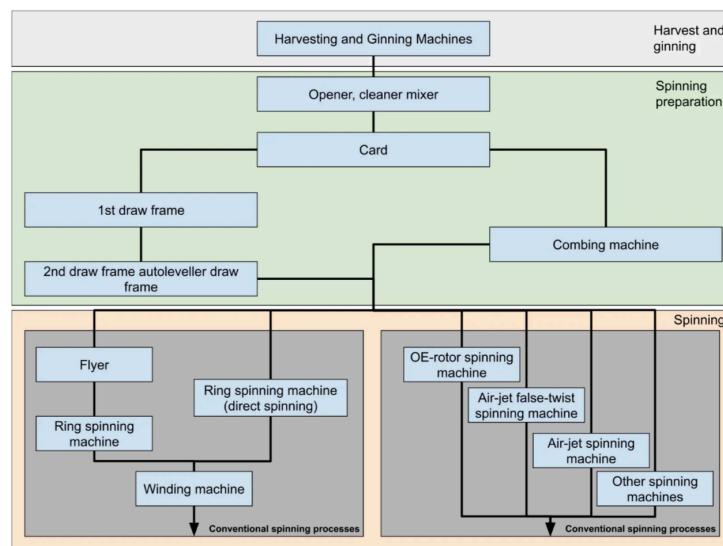




Spinning - Open-End (Rotor)

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1. Introduction



1.1 Purpose of the study

This document describes the life cycle inventory (LCI) model for open-end (rotor) spun yarn production as part of the Carbonfact Textile LCA Database. The aim is to provide a transparent, literature-based, and reproducible foreground inventory covering 14 fibre families across a continuous range of yarn fineness (40–600 dtex), yielding a grid of individually parameterised process datasets. The model is designed as a harmonised companion to the ring-spinning LCI: it shares the same capital goods convention, Input-Output Ratio (IOR) framework, plastic bobbin

modelling, andecoinvent activity naming, while incorporating electricity and lubrication parameters specific to the rotor technology.

Open-end (OE) rotor spinning is the second most widely deployed staple yarn technology globally, particularly dominant in the medium-to-coarse count segment (150–400 dtex) for denim, towelling, and casual knitwear. Its specific energy advantage over ring spinning, and the absence of the roving sub-process, make it the more energy-efficient choice for those count ranges — a distinction that this model makes quantitatively explicit across all 14 fibre families.

1.2 Spinning process — Open-End (Rotor)

In open-end rotor spinning, fibre preparation (opening, carding, drawing) is identical to ring spinning for cotton and most blends. The critical difference begins after drawing: the sliver is fed directly into the rotor box, eliminating the roving sub-process entirely. Inside the rotor box, fibres are individualised by the opening roller, transported by air current into the groove of a high-speed rotor (typically 80,000–150,000 rpm), and twisted by the rotor rotation as they are withdrawn through the navel and doffing tube. Yarn is wound continuously onto a package in the same integrated step.

Sub-process	Function	Comparison with ring spinning
Opening and cleaning	Loosens compressed bales; removes trash and short fibres	Identical to ring spinning for cotton and natural fibres
Carding	Produces a sliver from the opened fibre mass	Required for cotton and most fibres; typically one passage in OE vs up to two in ring
Drawing (single passage)	Drafts and doubles slivers to improve evenness	One passage in OE vs two (breaker + finisher) in ring
Open-end (rotor) spinning + winding	Opens fibres individually, transports into rotor, inserts twist, winds yarn continuously onto package	Replaces roving + ring frame + separate winder in ring spinning; the dominant SEC-determining step

The elimination of the roving stage and the higher mechanical efficiency of the rotor at medium counts explain the OE energy advantage, which narrows at fine counts where rotor speed limits and fibre cohesion become binding constraints.

1.3 System boundary

The system boundary covers gate-to-gate open-end spinning: from raw fibre input to finished yarn at the spinning mill gate. **Included:** direct and auxiliary electricity, lubricating oil input and waste mineral oil output, plastic bobbin input and waste polypropylene output, capital goods (machine amortisation and EoL steel), and building infrastructure. **Excluded:** fibre production (upstream); yarn losses and fibre waste (handled externally via the IOR — no inline yarn or textile waste row in any dataset); downstream processing; and **process water** (zero for all OE routes — OE spinning is a fully dry process).

1.4 Functional unit and scope

Parameter	Value
Functional unit	1 kg of finished OE yarn at the spinning mill gate
Background database	ecoinvent 3.12, system model: cut-off by classification
Background linking	Generic market activities, geography GLO or RoW
Impact method	IPCC 2021 GWP100 via Brightway2
Grid dimensions	14 fibre families × 47 dtex points (40–600 dtex)
dtex range	40–600 dtex (one step below the ring model, which starts at 45 dtex)
Process water	Zero for all OE routes — fully dry process
Harmonisation with ring model	Capital goods, plastic bobbins, ecoinvent activity names, and IOR convention are identical

2. Data sources and literature review

2.1 Review of existing LCI databases

Existing OE spinning datasets in EF 3.1 and ecoinvent 3.12 do not separate open-end from ring spinning at the technology level, or provide only single-point estimates without a fineness-resolved parameterisation. The EF 3.1 datasets for spinning typically report one electricity value for cotton at a reference count, without distinguishing the spinning technology. This absence is particularly notable given that OE spinning accounts for approximately 30% of global staple yarn production and operates at systematically different energy intensities from ring spinning. The Ecobalyse parametric model (ADEME, 2024) does distinguish OE from ring spinning at the formula level, providing a useful starting point, but its cotton baseline has not been validated against measured mill data. The primary motivation for this model is therefore to provide a technology-specific, empirically anchored, fineness-resolved OE spinning LCI.

2.2 Primary literature sources for open-end spinning electricity

2.2.1 Ecobalyse parametric model (ADEME, 2024)

Ecobalyse models OE spinning electricity as a function of yarn title in metric count (Nm), with a material constant $C_{OE} = 2$ kWh/kg at 50 Nm (versus $C_{ring} = 4$ for ring spinning):

$$SEC_{Ecobalyse,OE}(dtex) = \frac{400}{dtex} \quad [\text{kWh/kg}]$$

Key values: 40 dtex → 10.0 kWh/kg | 200 dtex → 2.0 kWh/kg | 600 dtex → 0.67 kWh/kg.

Average over 40–600 dtex: approximately 2.64 kWh/kg. The ratio to the ring spinning

Ecobalyse curve is exactly 0.5 at all counts, which is a structural simplification — empirical evidence suggests the OE/ring ratio is count-dependent, narrowing towards coarser yarns.

2.2.2 Kaplan & Koç (2010) — open-end rotor energy audit

Kaplan & Koç (2010) report a full-mill energy audit for carded OE cotton production, providing the most detailed measured energy breakdown for this technology available in the public literature. The key parameters from their Table 9 are:

Parameter	Value	Notes
Total energy for 3,000 kg of yarn	8,874.9 kWh	Includes machines + compressors + HVAC + lighting
Specific energy consumption at 200 dtex	2.95 kWh/kg	Full-mill SEC including all auxiliaries
ITMF range for same yarn type (1997–2006)	1.60–3.00 kWh/kg	Inter-mill variability; Kaplan value at upper end
OE rotor share of machine electricity	~75%	At 200 dtex, OE rotors dominate; opening/carding share is smaller

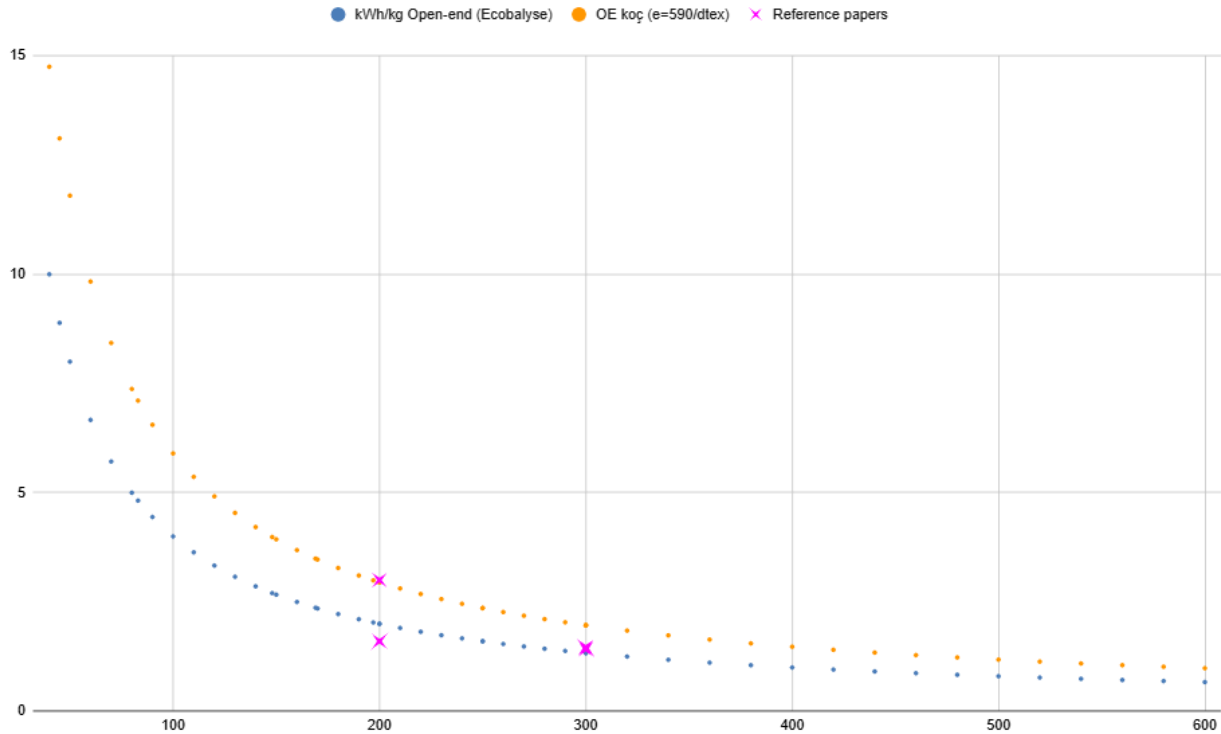
From this anchor point, an inverse-proportional curve is derived:

$$\text{SEC}_{\text{Kaplan,OE}}(\text{dtex}) = \frac{590}{\text{dtex}} \quad [\text{kWh/kg}]$$

Key values: 40 dtex → 14.75 kWh/kg | 200 dtex → 2.95 kWh/kg | 300 dtex → 1.97 kWh/kg | 600 dtex → 0.98 kWh/kg.

2.2.3 External validation anchor points

dtex	SEC (kWh/kg)	Source	Notes
200	1.60–3.00	ITMF (2010)	Country range for 20 tex OE cotton; confirms Kaplan value at upper bound
200	2.95	Kaplan & Koç (2010)	Primary anchor; full-mill with auxiliaries
300	1.42	van der Velden et al. (2014)	LCA benchmarking; cotton OE; likely narrower system boundary
300	1.46	ITMF (2010), mill-level average	Consistent with van der Velden

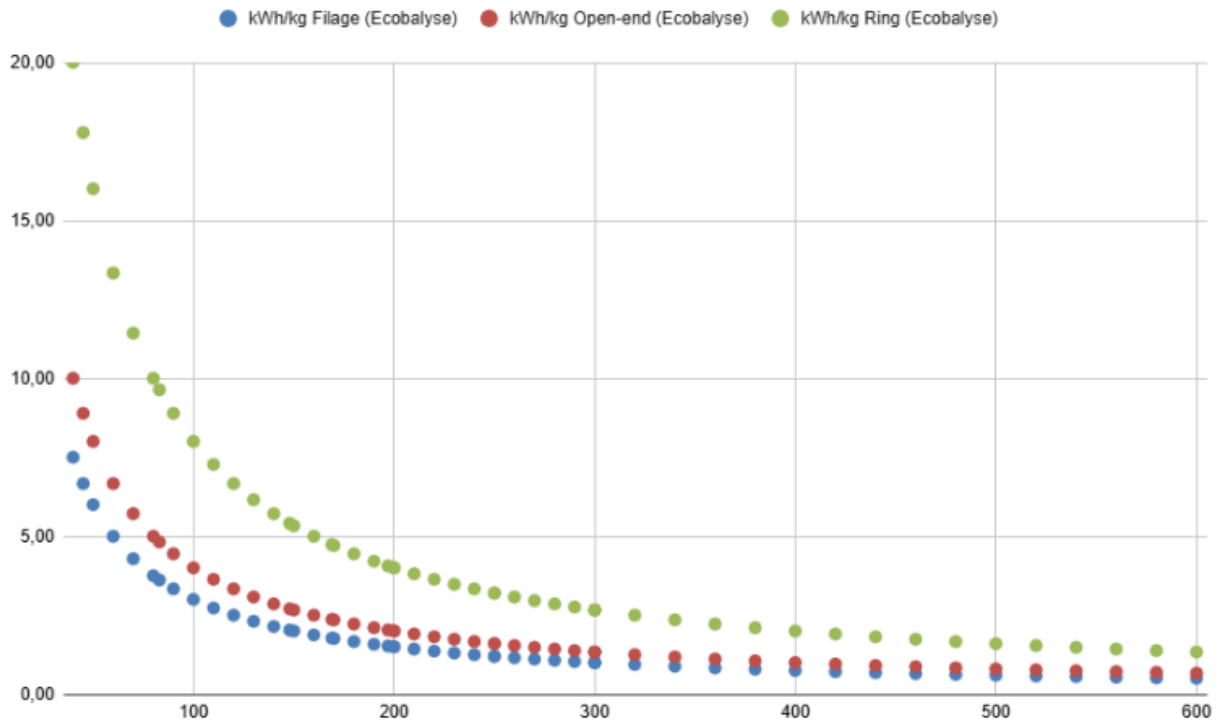


2.3 Comparison with ring spinning electricity

The OE rotor technology offers a systematic energy advantage over ring spinning across all dtex values, but the advantage narrows at coarser counts. The table below compares the blended OE average curve (500/dtex) against the ring spinning power-law from the companion document:

dtex	Ring SEC (kWh/kg)	OE average SEC (kWh/kg)	OE/Ring ratio
100	≈ 8.46	5.00	0.59
200	≈ 3.48	2.50	0.72
300	≈ 2.13	1.67	0.78
400	≈ 1.48	1.25	0.84
600	≈ 1.04	0.83	0.80

The narrowing OE advantage at coarser counts is consistent with the physical explanation: at coarser counts, both technologies operate at lower twist insertion rates per kg, and the roving elimination advantage of OE represents a smaller fraction of total energy. At fine counts, the ring spindle's efficiency advantage over the rotor opening roller becomes smaller, but the roving stage elimination remains dominant.



2.4 Sources for non-electricity parameters

- **Lubrication:** ITMF (2010) and Kaplan & Koç (2010) for cotton OE lubricant ranges; All/Made2Flow benchmark for fibre-family differentials; literature ranges from textile tribology sources.
- **Yarn losses (IOR):** Kaplan & Koç (2010); ITMF (2010); Altun (2012); Goyal & Nayak (2020); Jahid et al. (2020); Tausif et al. (2018); Akbaş & Çelik (2016); mill case studies.
- **Capital goods:** Identical to ring spinning model (see shared methodology page).
- **Plastic bobbins:** BREF (European Commission, 2003); same value as ring model (0.001 kg/kg).

3. Life cycle inventory modelling

3.1 Electricity model

3.1.1 Conceptual basis

As in ring spinning, OE rotor SEC scales inversely with yarn fineness. The physical mechanism is the same: finer yarns require more rotor rotations per kilogram. The inverse relationship is well-established in the OE literature. However, OE spinning differs from ring spinning in two important respects that affect the electricity model: (1) the absence of the roving stage removes

a sub-process whose energy is count-dependent; and (2) the rotor opening roller energy is largely independent of count, making the SEC curve somewhat shallower than the ring frame curve at fine counts.

3.1.2 Step 1 — Ecobalyse baseline for open-end rotor

$$SEC_{\text{Ecobalyse,OE}}(\text{dtex}) = \frac{400}{\text{dtex}} \quad [\text{kWh/kg}]$$

Derived from: $\$(Nm/50) \text{ times } 2 = (10\{, \}000/(\text{text{dtex}} \text{ times } 50)) \text{ times } 2 = 400/\text{text{dtex}}\$$.
Reference: ADEME & Ministère de la Transition Écologique (2024), $\$C_{\text{text{OE}}} = 2\$$.

3.1.3 Step 2 — Kaplan-based curve anchored to measured data

$$SEC_{\text{Kaplan,OE}}(\text{dtex}) = \frac{590}{\text{dtex}} \quad [\text{kWh/kg}]$$

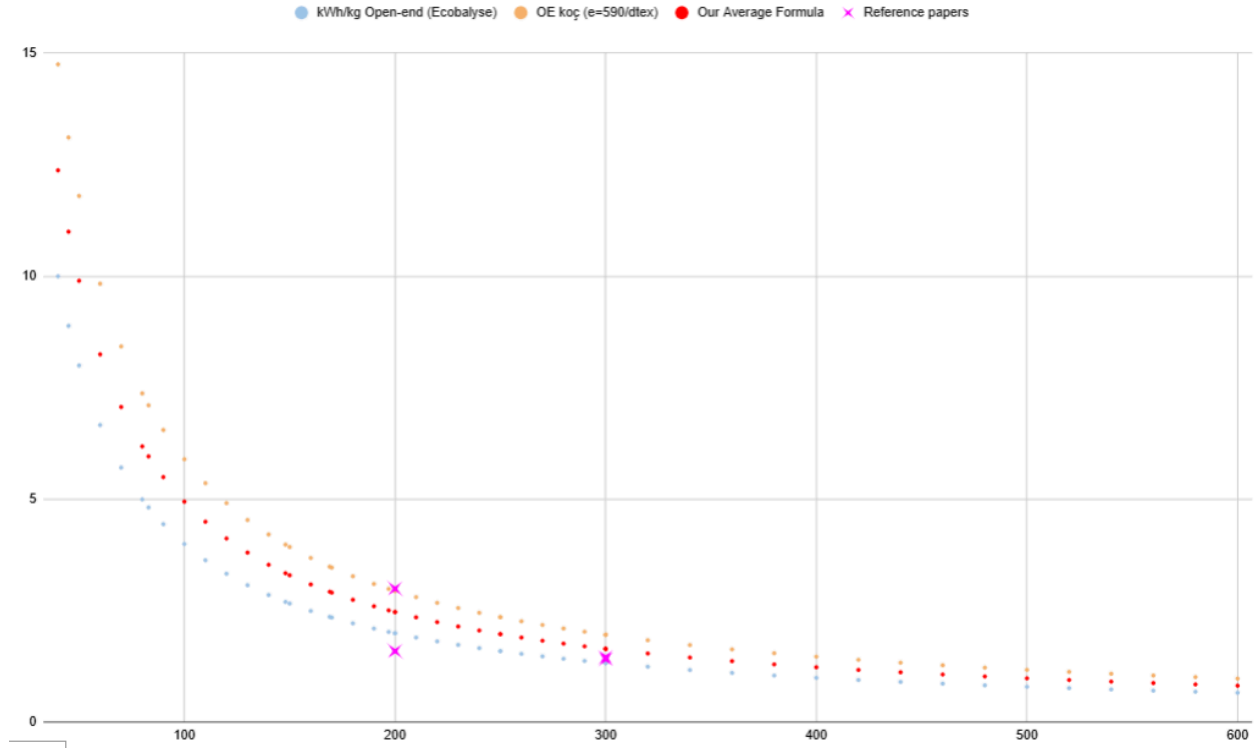
Anchor: 200 dtex → 2.95 kWh/kg (Kaplan & Koç, 2010, Table 9, full-mill including HVAC and lighting). Coarse anchor: ~490 dtex → ~1.20 kWh/kg (derived from Table 8, Kaplan & Koç, 2010). The ITMF range at 200 dtex (1.60–3.00 kWh/kg) confirms that the Kaplan value represents the upper end of the inter-mill range, reflecting full auxiliary inclusion.

3.1.4 Step 3 — Blended “Carbonfact OE average” baseline

The two sources are averaged to reduce single-source dependency:

$$SEC_{\text{CF,OE,avg}}(\text{dtex}) = \frac{SEC_{\text{Ecobalyse}} + SEC_{\text{Kaplan}}}{2} = \frac{400/\text{dtex} + 590/\text{dtex}}{2} = \frac{495}{\text{dtex}} \approx \boxed{\frac{500}{\text{dtex}}} \quad [\text{kWh/kg}]$$

Key values: 40 dtex → 12.5 kWh/kg | 100 dtex → 5.0 kWh/kg | 200 dtex → 2.5 kWh/kg | 300 dtex → 1.67 kWh/kg | 400 dtex → 1.25 kWh/kg | 600 dtex → 0.83 kWh/kg.



3.1.5 Step 4 — Implementation: cotton baseline and material factors

The blended average (500/dtex) is a cross-material reference level, not a cotton-specific curve. Material-type factors f_{material} are applied relative to this average baseline:

$$\text{SEC}_{\text{OE,material}}(\text{dtex}) = f_{\text{material}} \times \frac{500}{\text{dtex}} \quad [\text{kWh/kg}]$$

The cotton factor is 0.86 relative to the average, consistent with external validation:

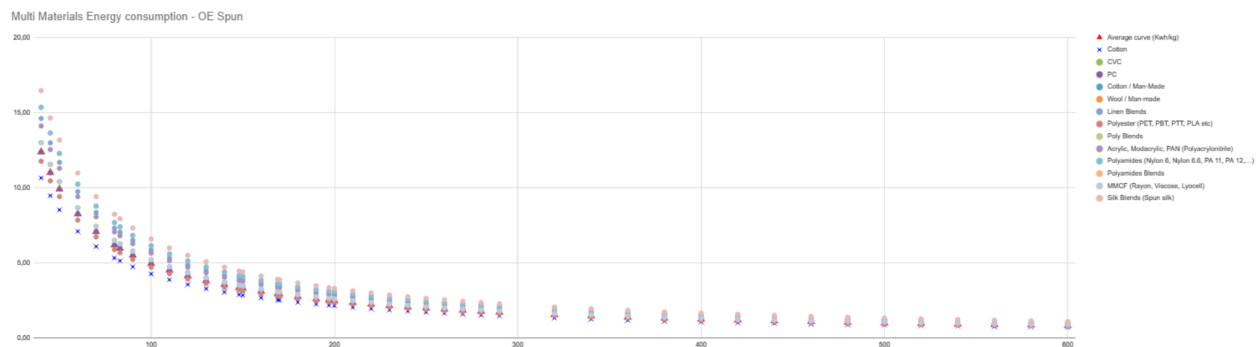
$$\text{SEC}_{\text{OE,cotton}}(\text{dtex}) = 0.86 \times \frac{500}{\text{dtex}} = \frac{430}{\text{dtex}} \quad [\text{kWh/kg}]$$

At 200 dtex: $0.86 \times 2.50 = 2.15$ kWh/kg (consistent with Kaplan 2.95 upper bound and ITMF 1.60 lower bound). At 300 dtex: $0.86 \times 1.67 = 1.43$ kWh/kg (consistent with van der Velden 1.42 — excellent match).

Material type	f_{material}	SEC at 200 dtex (kWh/kg)	Basis
Average baseline (CVC, PC)	1.00	2.50	Reference level; All/Made2Flow benchmarking
Cotton	0.86	2.15	All/Made2Flow; lower frictional demand in OE rotor
CVC (cotton-rich blends)	1.00	2.50	All/Made2Flow
PC (polyester–cotton 50/50)	1.00	2.50	All/Made2Flow

Material type	f_{material}	SEC at 200 dtex (kWh/kg)	Basis
Cotton / Man-Made (generic blends)	1.05	2.63	All/Made2Flow
Wool / Man-made blends	1.24	3.10	All/Made2Flow; wool fibre stiffness increases rotor load
Linen blends	1.18	2.95	All/Made2Flow; linen fibre stiffness and dust load
Polyester (PET, PBT, PTT, PLA)	0.95	2.38	All/Made2Flow; smooth synthetic fibre, lower rotor friction
Poly blends (generic synthetics)	1.05	2.63	All/Made2Flow
Acrylic, Modacrylic, PAN	1.14	2.85	All/Made2Flow; acrylic fibre static and friction
Polyamides (Nylon 6, 6.6, PA 11, PA 12)	1.24	3.10	All/Made2Flow
Polyamide blends	1.05	2.63	All/Made2Flow
MMCF (Rayon, Viscose, Lyocell)	1.05	2.63	All/Made2Flow
Silk blends (spun silk)	1.33	3.33	All/Made2Flow; lowest cohesion, highest break rate

⚠ Uncertainty note: The All/Made2Flow material-type factors are sourced from a single unpublished industry dataset (Apparel Impact Institute & Made2Flow, 2023) and cannot be independently verified. They are expressed relative to the 500/dtex average baseline, not relative to cotton: a factor of 0.86 for cotton means cotton consumes 14% less than the cross-material average. Treat as central estimates with **±20–30% uncertainty** until fibre-specific OE mill metering data become available.



3.1.6 Clamping rules

- dtex < 40: apply SEC at 40 dtex; document as extrapolation with ≥50% uncertainty

- dtex > 600: apply SEC at 600 dtex unless primary mill data are available
- Note: OE spinning is industrially uncommon below 150 dtex; ring spinning should be the primary modelling choice for fine counts

3.1.7 Uncertainty quantification

Zone	dtex range	Uncertainty band	Primary reason
Fine-count extrapolation	< 80 dtex	±30–50%	Very limited OE data at fine counts; OE technology rarely used below 150 dtex industrially
Well-anchored range	80–300 dtex	±15–25%	ITMF range and Kaplan provide empirical coverage; van der Velden validates at 300 dtex
Coarse-count extrapolation	300–600 dtex	±20–30%	Fewer independent sources; energy contribution is small but relative uncertainty remains
Non-cotton material factors	All dtex	Additional ±20–30%	Single unpublished All/Made2Flow source; no independent verification

3.2 Lubrication model

3.2.1 Literature-informed ranges and production values

Lubricant consumption in OE rotor spinning follows a decreasing function of dtex for all fibre families, reflecting the higher surface contact frequency at finer counts. In contrast to ring spinning, OE rotor spinning applies lubricant primarily to the rotor and opening roller components rather than to ring travellers, resulting in systematically different consumption profiles by fibre type. The ranges below are derived from the literature cited; the production value at 200 dtex represents the mid-point of the implemented range.

Fibre family	Range (kg/kg)	Prod. value 200 dtex (kg/kg)	Observation	References
Cotton	0.0005–0.0015	0.00129	Synthetic fibres generate higher friction in rotor zone due to smoother surface; cotton is the lower-demand baseline	Lawrence & Chen (1984); Hove (1990)
CVC	0.0010–0.0020	0.00179	Lubricants must be adjusted based on synthetic blend ratio to minimise fibre-to-metal friction	Elhawary (2015); Kaplan (2010)

Fibre family	Range (kg/kg)	Prod. value 200 dtex (kg/kg)	Observation	References
PC (polyester-cotton 50/50)	0.0015–0.0030	0.00272	Greater reduction in losses with PET but increased lint generation; sensitive to lubrication	Elhawary (2015); Goyal & Nayak (2020); Kaplan (2010)
Cotton / Man-Made	0.0015–0.0030	0.00272	Synthetic fibres require additional lubricant and antistatic agents to ensure smooth separation	Ishtiaque & Kothari (2003); Hu et al. (2020)
Wool / Man-made blends	0.0010–0.0025	0.00222	Lubricants must be adjusted based on synthetic blend ratio to minimise fibre-to-metal friction	Elhawary (2015)
Linen blends	0.0010–0.0025	0.00222	Lubricants must be adjusted based on synthetic blend ratio to minimise fibre-to-metal friction	Elhawary (2015)
Polyester (PET, PBT, PTT, PLA)	0.0020–0.0040	0.00338	Synthetic fibres require additional lubricant and antistatic agents to ensure smooth separation	Ishtiaque & Kothari (2003); Hu et al. (2020)
Poly blends (generic synthetics)	0.0020–0.0035	0.00299	Lubricants must be adjusted based on synthetic blend ratio	Elhawary (2015)
Acrylic, Modacrylic, PAN	0.0020–0.0040	0.00338	Generates more dust without proper lubrication; requires additional antistatic agents	Ishtiaque & Kothari (2003); Hu et al. (2020)
Polyamides (Nylon 6, 6.6, PA 11, PA 12)	0.0020–0.0040	0.00338	Requires additional lubricant and antistatic agents; rotor cooling needed	Ishtiaque & Kothari (2003); Hu et al. (2020)
Polyamide blends	0.0020–0.0040	0.00338	Synthetic fibres require additional lubricant and antistatic agents to ensure smooth separation	Ishtiaque & Kothari (2003); Hu et al. (2020)
MMCF (Rayon, Viscose, Lyocell)	0.0015–0.0025	0.00229	Lyocell fibres benefit from lubrication to reduce rotor wear; regenerated fibres	Tausif et al. (2018)

Fibre family	Range (kg/kg)	Prod. value 200 dtex (kg/kg)	Observation	References
			sensitive to moisture and electrostatic charges	
Silk blends (spun silk)	0.0010–0.0020	0.00179	Low cohesion and high break rate in rotor despite low lubricant demand	Akbaş & Çelik (2016); Hu et al. (2020)

3.3 Process water

Process water consumption is **zero for all OE fibre families**. Open-end rotor spinning is a fully dry process: there are no wet preparation, scouring, or conditioning steps in the spinning hall. This distinguishes OE spinning from the ring model, where MMCF and wool routes carry non-zero water consumption. The absence of process water in OE is treated as a structural model property, not a default assumption requiring future revision.

3.4 Yarn loss and Input–Output Ratio (IOR)

Yarn waste and losses per kilogram of finished OE yarn are not modelled inline. They are communicated to the caller via the Input–Output Ratio (IOR):

$$IOR = \frac{1}{1 - L}$$

where L is the fibre/yarn loss rate as a fraction. For every 1 kg of finished OE yarn, the caller supplies $IOR \times 1$ kg of upstream fibre. The lost fraction ($IOR - 1$) kg is routed to **market for waste yarn and waste textile {RoW} | cut-off**.

Example: Cotton at 200 dtex: $L = 0.167$ $IOR = 1/(1-0.167) = 1.200$.

⚠ Note on industrial scope: Not all fibre families covered by this model are routinely processed on OE rotor machines in practice. Elastane (Spandex/Lycra) is almost exclusively ring-spun. Fine-count yarns (< 150 dtex) are technically possible on OE but uncommon commercially. The OE model is most robust for the 150–400 dtex range.

Representative IOR values at 200 dtex:

Material / Route	Literature range (loss %)	Loss at 200 dtex (%)	IOR (kg/kg)	Observation	References
Cotton	6.5–19.2%	16.67%	1.200	Losses vary with type (carded vs combed); husks and short fibres increase rejection;	Gabriela et al. (2025); Altun (2012); Kaplan (2010); ITMF (2010)

Material / Route	Literature range (loss %)	Loss at 200 dtex (%)	IOR (kg/kg)	Observation	References
				opening/carding + rotor residue	
CVC	11–15%	13.63%	1.158	Blending with PES helps consistency but cotton impurities still cause losses; PES improves regularity	Kaplan (2010); Hu et al. (2020)
PC (polyester–cotton 50/50)	9–14%	12.91%	1.148	Greater reduction with PET but increased lint generation; sensitive to lubrication	Goyal & Nayak (2020); Kaplan (2010)
Cotton / Man-Made	12–16%	14.91%	1.175	Performance depends on proportion and type of synthetic; rotor requires uniformity	Hu et al. (2020)
Wool / Man-made blends	18–25%	23.08%	1.300	OE requires short fibres; wool has low cohesion; high residue; difficult reliability	Hu et al. (2020)
Linen blends	15–20%	18.63%	1.229	Long, loosely bound fibres cause clogging and rotor breakage; high discards	Hu et al. (2020)
Polyester (PET, PBT, PTT, PLA)	3–5%	4.45%	1.047	OE runs well with pure synthetic fibres; very low and consistent losses	Altun (2012); Goyal & Nayak (2020)

Material / Route	Literature range (loss %)	Loss at 200 dtex (%)	IOR (kg/kg)	Observation	References
Poly blends (generic synthetics)	7–10%	9.17%	1.101	More uniform the mixture, lower the losses; acrylic or PLA can increase loss	Hu et al. (2020); Goyal & Nayak (2020)
Acrylic, Modacrylic, PAN	10–13%	12.17%	1.139	Good compatibility with OE but generates more dust; average losses without proper lubrication	Altun (2012); Jahid et al. (2020)
Polyamides (Nylon 6, 6.6, PA 11, PA 12)	3–5%	4.45%	1.047	High regularity allows low losses; requires rotor cooling	Altun (2012); Hu et al. (2020)
Polyamide blends	8–10%	9.45%	1.104	Nylon mixed with natural fibres loses uniformity and can cause more breakage	Hu et al. (2020)
MMCF (Rayon, Viscose, Lyocell)	9–12%	11.17%	1.126	Regenerated fibres perform well but are sensitive to moisture and electrostatic charges	Goyal & Nayak (2020); Tausif et al. (2018); Kaplan (2010)
Silk blends (spun silk)	16–24%	21.82%	1.279	Unstable spinning; low cohesion and rotor breaks; very high losses even with blends	Akbaş & Çelik (2016)

The full IOR table covering all dtex combinations is maintained in [Spinning_IORvalues.xlsx](#).

3.5 Capital goods: machinery and buildings

$$\text{Amount (unit/kg)} = \frac{1}{Q_{\text{lifetime}}} = \frac{1}{2,155,172} \approx \boxed{4.64 \times 10^{-7} \text{ unit/kg}}$$

$$Q_{\text{lifetime}} = 15 \text{ yr} \times 300 \text{ days/yr} \times 16 \text{ h/day} \times \text{throughput} = 2,155,172 \text{ kg}$$

EoL steel: 0.0013 kg/kg →

market for waste steel {RoW} | waste steel | cut-off

Unlike the ring spinning model, open-end spinning uses a **single standard capital goods group** for all fibre families. There is no combing-line differentiation because OE spinning does not include a combing preparation stage.

Shared methodology: Capital Goods & End-of-Life (Machinery) in LCA — Full derivation, lifetime parameters, and EoL split. Process-specific value for OE spinning: **4.64×10^{-7} unit/kg** for all OE variants, EoL steel **0.0013 kg/kg**.

Shared methodology: Building Infrastructure in LCA — Process-specific value for OE spinning: **$8.84 \times 10^{-6} \text{ m}^2/\text{kg}$** (cf. ring model: $1.0 \times 10^{-5} \text{ m}^2/\text{kg}$), constant across all OE variants. Activity: **building, hall {GL0} | building, hall | cut-off**.

3.6 Plastic bobbins

Plastic carriers (bobbins) are modelled identically to the ring spinning model: **0.001 kg of polypropylene per kilogram of yarn** → **market for polypropylene, granulate {GL0} | polypropylene, granulate | cut-off**. Equal mass output as waste polypropylene → **waste polypropylene, for recycling, unsorted {GL0} | Recycled Content cut-off**. Constant across all fibre families and dtex.

3.7 Dataset naming convention

All OE datasets follow: **Spinning to yarn, [Fibre family], [dtex] dtex, Open-End (Rotor)**. Example: **Spinning to yarn, Cotton, 200 dtex, Open-End (Rotor)**.

4. Life cycle impact assessment

4.1 LCIA results

Climate change impacts (kg CO₂eq / kg yarn) were computed using Brightway2 with the IPCC 2021 GWP100 characterisation factors, applied to the foreground inventory with ecoinvent 3.12 (cut-off) background processes and the GLO average electricity mix. The dataset spans 54 process variants across 6 fibre families and 9 dtex points (45–370 dtex); a representative selection is shown below. Full results across all 16 EF 3.1 impact indicators are available in **impact-scores.csv**.

Cotton — full dtex range

Process	GHG (kg CO ₂ eq / kg yarn)
Spinning to yarn, Cotton, 45 dtex, Open-End (Rotor)	7.32
Spinning to yarn, Cotton, 70 dtex, Open-End (Rotor)	4.89
Spinning to yarn, Cotton, 120 dtex, Open-End (Rotor)	2.68
Spinning to yarn, Cotton, 200 dtex, Open-End (Rotor)	1.51
Spinning to yarn, Cotton, 300 dtex, Open-End (Rotor)	1.03
Spinning to yarn, Cotton, 370 dtex, Open-End (Rotor)	0.814

Synthetic and man-made fibres — 200 dtex reference count

Process	GHG (kg CO ₂ eq / kg yarn)
Spinning to yarn, Acrylic, Modacrylic, PAN, 200 dtex	2.00
Spinning to yarn, MMCF (Rayon, Viscose, Lyocell), 200 dtex	1.84
Spinning to yarn, Polyamides (Nylon 6, 6.6, PA 11, PA 12, ...), 200 dtex	2.18
Spinning to yarn, Polyester (PET, PBT, PTT, PLA etc.), 200 dtex	1.67

Silk — highest-GHG OE material family, full dtex range

dtex	GHG (kg CO ₂ eq / kg yarn)
45	11.31
70	7.55
120	4.14
200	2.32
300	1.58
370	1.25

Results confirm the expected inverse relationship between yarn fineness and GHG intensity. The OE values are systematically lower than ring spinning equivalents at all counts: at 200 dtex, cotton OE (1.51 kg CO₂eq/kg) is approximately 38% lower than cotton combed knitting ring (2.45 kg CO₂eq/kg), consistent with the 0.72 OE/ring SEC ratio at that count. The GLO average electricity mix is the primary driver; substituting a regional grid shifts absolute values while preserving relative rankings.

4.2 Comparison with EF 3.1 and ring spinning

The Carbonfact OE spinning model differs from EF 3.1 in four structural ways. First, it provides a continuous SEC function of dtex for 14 fibre families, whereas EF 3.1 does not distinguish OE from ring spinning at the dataset level. Second, it applies material-type factors explicitly calibrated to OE rotor technology (rather than inheriting ring spinning factors). Third, it documents OE-specific IOR values per material family, which are systematically lower than ring spinning IORs due to the absence of a combing stage for cotton. Fourth, it applies an OE-specific building intensity ($8.84 \times 10^{-6} \text{ m}^2/\text{kg}$) slightly lower than the ring model ($1.0 \times 10^{-5} \text{ m}^2/\text{kg}$), reflecting the more compact layout of OE spinning lines.

5. Inferences and limitations

The electricity model is based on two sources (Ecobalyse and Kaplan & Koç, 2010) and their simple average. While the cotton baseline is well-validated at 200 and 300 dtex, there are no independent measured anchor points below 80 dtex in the OE literature, because OE spinning is rarely used at fine counts commercially. The fine-count portion of the model (below 80 dtex) should be treated as indicative and ring spinning should be used as the preferred technology model for those ranges.

The material-type SEC factors (Section 3.1.5) are derived from the same unpublished All/Made2Flow dataset used in the ring model. They have been re-normalised to the OE average baseline (500/dtex) rather than to the ring cotton baseline, but they carry the same $\pm 20\text{--}30\%$ uncertainty and have not been independently verified for OE technology specifically. Priority should be given to collecting OE mill metering data for acrylic, polyamide, and MMCF routes, where the factors carry the most uncertainty.

The IOR values for non-cotton fibres in OE spinning are less well-documented in the published literature than their ring spinning equivalents, because OE research has historically focused on cotton. The values for polyester and polyamide (4.45%) are sourced from Goyal & Nayak (2020) and are considered reliable; the values for silk blends (21.82%), wool/man-made blends (23.08%), and linen blends (18.63%) are derived from mill case studies and carry higher uncertainty ($\pm 30\text{--}50\%$). Process water is zero for all OE routes and this is a structural property of the technology, not a gap.

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7. Annex

LCI for 1 kg of open-end yarn — Cotton, 200 dtex (representative example)

Flow	Value	Unit	Source / Rationale	
Electricity (input)	2.16	kWh	$SEC_{OE,cotton}(200) = 0.86 \times (500/200) = 2.15 \text{ kWh/kg}$ (production file: 2.16 kWh/kg). `market group for electricity, medium voltage {GLO} \	electricity, medium voltage \
Lubricating oil (input)	0.00129	kg	Cotton OE lubricant at 200 dtex; decreasing profile. `lubricating oil production {RoW} \	lubricating oil \
Waste mineral oil (output)	0.00129	kg	Equal to lubricant input. `market for waste mineral oil {RoW} \	waste mineral oil \
Polypropylene bobbin (input)	0.001	kg	Constant, all variants. `market for polypropylene, granulate {GLO} \	polypropylene, granulate \
Waste PP bobbin (output)	0.001	kg	`waste polypropylene, for recycling, unsorted {GLO} \	Recycled Content cut-off`
Capital goods – machine (input)	4.64×10^{-7}	unit	Single standard variant for all OE fibre families. `market for building machine {GLO} \	building machine \
EoL machine – steel (output)	0.0013	kg	`market for waste steel {RoW} \	waste steel \
Building infrastructure (input)	8.84×10^{-6}	m ²	OE-specific (cf. ring: $1.0 \times 10^{-5} \text{ m}^2/\text{kg}$). `building, hall {GLO} \	building, hall \
Process water	0	kg	Zero for all OE routes — fully dry process.	
Yarn / fibre input	— (IOR)	kg	IOR = 1.200 at 200 dtex (cotton OE loss = 16.67%). Caller supplies 1.200 kg upstream fibre; lost fraction 0.200 kg/kg → `market for waste yarn and waste textile {RoW} \	cut-off`.

LCI for 1 kg of open-end yarn — Acrylic, Modacrylic, PAN, 200 dtex

Flow	Value	Unit	Source / Rationale	
Electricity (input)	2.86	kWh	$SEC_{OE,acrylic}(200) = 1.14 \times (500/200) = 2.85 \text{ kWh/kg}$ (production file: 2.86 kWh/kg). `market group for electricity, medium voltage {GLO} \	cut-off`
Lubricating oil (input)	0.00338	kg	Higher friction synthetic fibre; decreasing with dtex. `lubricating oil production {RoW} \	cut-off`
Waste mineral oil (output)	0.00338	kg	`market for waste mineral oil {RoW} \	cut-off`
Polypropylene bobbin (input)	0.001	kg	`market for polypropylene, granulate {GLO} \	cut-off`
Waste PP bobbin (output)	0.001	kg	`waste polypropylene, for recycling, unsorted {GLO} \	Recycled Content cut-off`
Capital goods – machine (input)	4.64×10^{-7}	unit	`market for building machine {GLO} \	cut-off`
EoL machine – steel (output)	0.0013	kg	`market for waste steel {RoW} \	cut-off`
Building infrastructure (input)	8.84×10^{-6}	m ²	`building, hall {GLO} \	cut-off`
Process water	0	kg	Zero — dry process.	
Yarn / fibre input	— (IOR)	kg	IOR = 1.139 at 200 dtex (acrylic OE loss = 12.17%). Lost fraction → `market for waste yarn and waste textile {RoW} \	cut-off`.

LCI for 1 kg of open-end yarn — Silk Blends (Spun silk), 200 dtex

Flow	Value	Unit	Source / Rationale	
Electricity (input)	3.34	kWh	$SEC_{OE,silk}(200) = 1.33 \times (500/200) = 3.33 \text{ kWh/kg}$ (production file: 3.34 kWh/kg). `market group for electricity, medium voltage {GLO} \	cut-off`

Flow	Value	Unit	Source / Rationale	
Lubricating oil (input)	0.00179	kg	Lower fibre-to-metal friction despite high break rate; decreasing with dtex. `lubricating oil production {RoW} \	cut-off`
Waste mineral oil (output)	0.00179	kg	`market for waste mineral oil {RoW} \	cut-off`
Polypropylene bobbin (input)	0.001	kg	`market for polypropylene, granulate {GLO} \	cut-off`
Waste PP bobbin (output)	0.001	kg	`waste polypropylene, for recycling, unsorted {GLO} \	Recycled Content cut-off`
Capital goods – machine (input)	4.64×10^{-7}	unit	`market for building machine {GLO} \	cut-off`
EoL machine – steel (output)	0.0013	kg	`market for waste steel {RoW} \	cut-off`
Building infrastructure (input)	8.84×10^{-6}	m ²	`building, hall {GLO} \	cut-off`
Process water	0	kg	Zero — dry process.	
Yarn / fibre input	— (IOR)	kg	IOR = 1.279 at 200 dtex (silk blend OE loss = 21.82%). Lost fraction → `market for waste yarn and waste textile {RoW} \	cut-off`.

IOR table — all OE fibre families at 200 dtex

Material / Route	Loss (%)	IOR (kg/kg)	Literature range	References
Cotton	16.67%	1.200	6.5–19.2%	Gabriela et al. (2025); Altun (2012); Kaplan (2010); ITMF (2010)
CVC (cotton-rich blends)	13.63%	1.158	11–15%	Kaplan (2010); Hu et al. (2020)
PC (polyester–cotton 50/50)	12.91%	1.148	9–14%	Goyal & Nayak (2020); Kaplan (2010)
Cotton / Man-Made (generic blends)	14.91%	1.175	12–16%	Hu et al. (2020)
Wool / Man-made blends	23.08%	1.300	18–25%	Hu et al. (2020)
Linen blends	18.63%	1.229	15–20%	Hu et al. (2020)

Material / Route	Loss (%)	IOR (kg/kg)	Literature range	References
Polyester (PET, PBT, PTT, PLA)	4.45%	1.047	3–5%	Altun (2012); Goyal & Nayak (2020)
Poly blends	9.17%	1.101	7–10%	Hu et al. (2020); Goyal & Nayak (2020)
Acrylic, Modacrylic, PAN	12.17%	1.139	10–13%	Altun (2012); Jahid et al. (2020)
Polyamides	4.45%	1.047	3–5%	Altun (2012); Hu et al. (2020)
Polyamide blends	9.45%	1.104	8–10%	Hu et al. (2020)
MMCF (Rayon, Viscose, Lyocell)	11.17%	1.126	9–12%	Goyal & Nayak (2020); Tausif et al. (2018); Kaplan (2010)
Silk blends (spun silk)	21.82%	1.279	16–24%	Akbaş & Çelik (2016)